

Hao Zhang

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EDUCATION

University of California, Santa Barbara

California, USA

Major Courses: PSTAT 231 Introduction to Statistical Machine Learning, Math 104C Advanced Topics in Numerical Analysis

GPA: 4.0/4.0

Mar 2026 – Jun 2026

University of California, Berkeley

California, USA

Major Courses: Math 254A Algebraic Number Theory, Math 250A Groups, Rings and Fields, Math 250B Commutative Algebra, Math 128B Numerical Analysis

GPA: 3.3/4.0

Jan 2025 – Dec 2025

Dalian University of Technology

Dalian, China

Bachelor of Science in Mathematics

Sep 2022 – Jan 2025

Major Courses: Mathematical Analysis H, Elementary Number Theory, Complex Function Theory H, Geometry H, Abstract Algebra H, Ordinary Differential Equations H, Theory of Functions of Real Variables H, Probability and Mathematical Statistics H, Complex Representation of Finite Groups, Optimization Method H, Numerical Approximation H, Elementary Topology H

Note: Courses marked with “H” are honors-level, indicating increased difficulty and academic rigor.

RESEARCH EXPERIENCES

Log-Free Hyperelliptic Specialization and Class Groups of Quadratic Fields

Preprint, 2026

Westlake University, Hangzhou, China

Research Assistant, Advisor: Prof. Yue Zhang

Jun 2026 – Sep 2026

- Established a log-free hyperelliptic specialization theorem for quadratic class groups under a branch-orbit condition, removing the logarithmic loss in the Kulkarni–Levin bound; as an application, improved the lower bound for imaginary quadratic fields with class-group 5-rank at least two from $\frac{X^{1/3}}{(\log X)^2}$ to $X^{1/3}$.

Research Seminar on In-Context Learning, Transformer Theory, and Large-Model Optimization

UCSB, California, USA

Research Assistant, Advisor: Prof. Sui Tang

Mar 2026 – Jun 2026

- Developed and analyzed causal kernel descent (CKD) for two-lag prediction $x_{t+1} = g(x_t, x_{t-1})$, introducing direct pair-kernel and lifted-state formulations for in-context autoregressive learning.
- Proved exact finite-depth equivalence and residual-convergence guarantees for the two formulations, and characterized a nontrivial family of linear two-lag systems admitting a shared invariant metric and CKD parameterization.

Residual-Level Convolution-Guided Adaptive Resampling for Physics-Informed SnowSea-Ice Heat Transfer Modeling

UCSB, California, USA

Research Assistant, Advisor: Prof. Guo Yu

Mar 2026 – Jun 2026

- Developed a physics-informed snow–sea-ice heat-transfer model with solar-radiation forcing and snow-layer thickness, and designed convolution-guided adaptive resampling based on smoothed PDE-residual maps.
- Improved over the fixed-weight PINN baseline, reducing RMSE from 1.5221 to 1.2317 and MAE from 1.0138 to 0.7025, while increasing R^2 from 0.9240 to 0.9503.

Expository Paper on Lubin–Tate Theory

UCB, California, USA

Independent Researcher, Advisor: Prof. Sugwoo Shin, Prof. Xiaolei Liu

Aug 2025 – Dec 2025

- Completed an expository thesis on Lubin–Tate theory and the explicit construction of local class field theory via formal groups and Lubin–Tate extensions.

HONORS & AWARDS

National Second Prize, Chinese Mathematical Competition (top 12% out of 325,304 students) Dec 2024

National Third Prize, Chinese Mathematical Competition (top 15% out of 288,355 students) Dec 2023

Advanced to the Preliminary Round of the Alibaba Global Mathematics Competition (top 0.32% of 250,000+ participants) Jun 2024

Science and Innovation Scholarship, Dalian University of Technology (top 4% of 180 students) Aug 2024

Innovation Scholarship, School of Mathematical Sciences, Dalian University of Technology Jan 2023